Figure 19.1 shows the fraction of all electricity generated by nuclear power stations in different countries. In France any further large increase in nuclear power production capacity is unlikely as this could lead to a generating capacity in excess of demand. Furthermore, opposition to the peaceful use of nuclear power has led to revised plans for the use of nuclear power in many countries in the West (cf. Ch. 22). In some cases earlier expansion plans have been replaced by moratoria on new plants, by a decision to shut-down some or all operating nuclear power stations (Germany, Italy, and Lithuania), or by plans to force the utilities to shut down all nuclear power stations at some future date (Germany).

Nuclear chemistry plays an essential part in achieving safety and reliability in this source of power. Nuclear chemists and engineers are responsible for much of the nuclear fuel cycle, from uranium ore processing to ultimate disposal of radioactive waste.

In the future fusion power may also contribute to the energy supply. However, not all problems have been overcome to date, see §19.17.

19.1. THE NUCLEAR REACTOR

The fission process with thermal neutrons can be summarized as follows

$$^{235}\mathrm{U} + \mathrm{n} \rightarrow \mathrm{FP} + \nu\mathrm{n} \tag{19.1}$$

For each neutron consumed on the average ~ 2.5 ($= \nu$) new neutrons are released. The new neutrons can be used to fission other ^{235}U nuclei leading to the release of even more neutrons forming a *nuclear chain reacting system*. The kinetic energy of the fission fragments is promptly converted to heat through collisions with other atoms. In a *nuclear reactor* the nuclear chain reactions are controlled so that an equilibrium state is reached, where for each fission exactly one of the new neutrons is used for further fission. Under these conditions the *neutron multiplication factor k* is said to be 1. If the factor is higher the number of neutrons, and consequently the fission rate, increases exponentially. Without any control mechanism the heat evolved would ultimately destroy the chain-reacting system. This is extremely improbable in conventional reactors because of various active and passive control mechanisms. No such control mechanisms are present in *nuclear explosive devices* and, instead, every effort is made to make the reaction as violent as possible.

The neutron transport equation describes the interaction of neutrons with their environment. The accurate design of a chain reacting system requires solution of this equation. The complete solution yields the neutron flux and neutron spectrum as function of the space coordinates and time. As there is no known analytical solution to this equation for a real chain reacting system, numerical methods or various approximations must be used. In this text we do not consider numerical solution methods. We illustrate the general properties of chain reacting systems of interest to the nuclear

chemist by using some approximate solutions assuming that the neutron flux and its energy spectrum is known.

There is quite a flexibility in the design of a controlled nuclear chain reacting system. Because each concept has its advantages and drawbacks, more than a dozen different types of nuclear reactors have been developed and tested. However, only a few types are common. About 79% of the currently operating nuclear power plants are of the *light water reactor* type (LWR), i.e. *pressurized water reactors* (PWR)[3] or *boiling water reactors* (BWR); cf. Table 19.1. We therefore begin our discussion of reactors by describing a LWR station with emphasis on the principles and main components. Figure 19.2 shows a schematic diagram of a PWR and a picture of operating LWRs, three PWRs and one BWR (highest chimney).

The principal component in any reactor is the *core* which contains the fissionable *fuel material*. This is usually UO_2 enriched in ^{235}U to 2–4% shaped as short cylinders, *pellets*, which are stacked in zirconium alloy tubes (the *can*, *canning* or *cladding*) forming *fuel rods* or *fuel pins*. The fuel rods are mounted together in clusters forming *fuel elements* or *fuel assemblies*.

When fission occurs in the nuclear fuel practically all of the immediately released fission energy occurs as kinetic energy of fission fragments and neutrons (cf. Table 11.1 and §19.2). The kinetic energy is rapidly converted into heat. In order to maintain a stable operating temperature in the reactor it is necessary to provide a *coolant*. The coolant material, which is water in LWRs, enters the core from below, is heated while passing upwards along the fuel rods and leaves the core at the top as either hot water at high pressure (in the *pressurized water reactor*, PWR, as indicated in Figure 19.2) or as a

Table 19.1 Power reactor types in common use

Reactor type and abbreviation	Number of reactors (%)	Generating capacity (% of nuclear capacity)
Pressurized Water Reactor (PWR)	62.6	67.6
Boiling Water Reactor (BWR)	18.7	19.7
Pressurized Heavy Water Reactor (PHWR)	10.3	6.1
Pressure Tube Boiling Water Reactor (RBMK)	2.5	2.7
Advanced Gas-cooled Reactor (AGR)	3.2	2.3
Advanced Boiling Water Reactor (ABWR)	0.7	1.1
Magnox Reactor (GCR)	0.7	0.3
Fast Breeder Reactor (FBR)	0.5	0.2
Electricity Generating District Heating Reactor (EGP)	0.9	~0.01

Data refer to mars 2012.

[3] Until Ch.20 we will not discuss other existing pressurized water (or boiling water) reactor designs.

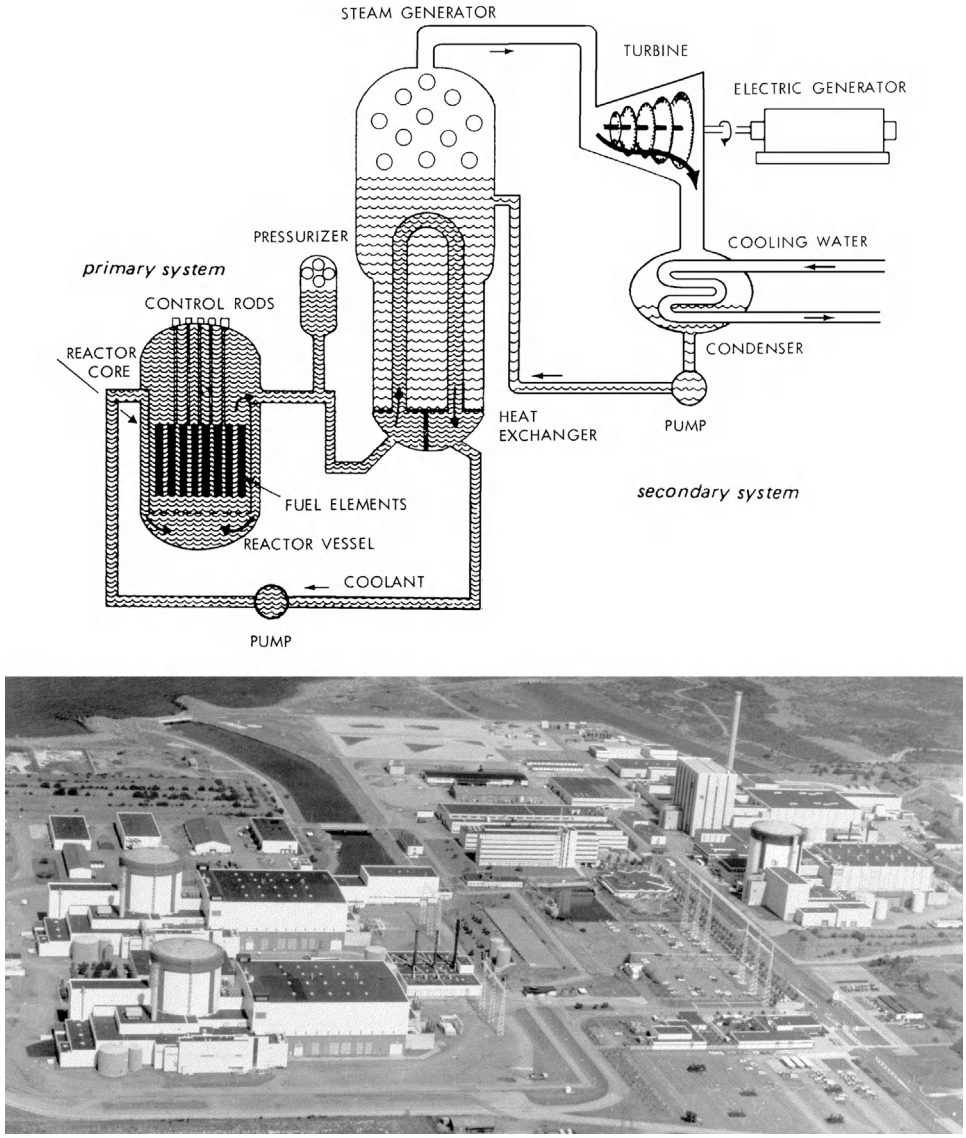


Figure 19.2 Main components of a pressurized light water cooled and -moderated nuclear power reactor (PWR) and a view of the Ringhals plant (Sweden) with 3 PWRs and 1 BWR.

high pressure high temperature steam – water mixture (in the *boiling water reactor*, BWR). The following discussion will concentrate on the operation of a PWR because it is the most common reactor type.

In the PWR steam is generated at lower pressure on the secondary side of separate heat exchangers, *steam generators*. The steam drives one or more turbines connected to

electric generators and is condensed to water by an *external cooling system* after the turbine. Except for the external cooling system (sea or river water, cooling towers, etc.), the steam-water flow systems are closed cycles.

The neutron flux, and hence the rate of the fission chain reaction, in the reactor core is controlled by movable *control rods* which contain material with high cross sections for absorption of neutrons and, in a PWR, also by an absorber dissolved in the coolant.

In order to maximize the cross section for fission, which is greatest for low energy neutrons, the neutrons are slowed down or "moderated" by a material (the *moderator*) that elastically scatters neutrons but has a small neutron capture cross section. In LWRs ordinary (but very pure) water serves the purpose of both moderation and cooling (in other reactor types the moderator may be a liquid like D_2O, a solid material like graphite or absent and the cooling medium may be a gas, e.g., helium or carbon dioxide or a metal like lead, mercury or sodium).

Reactors of this kind, in which the fuel is physically separated from the moderator, are said to be of the *heterogeneous type*, while in *homogeneous reactors* the fuel is directly dissolved in the moderator material. Homogeneous reactors have only been built for experimental purposes.

The reactor core system is enclosed in a stainless steel clad pressure vessel of high strength steel, thermally insulated on the outside. In order to protect the operating personnel against hazards from the neutrons and γ-rays emitted in fission, the pressure vessel is surrounded by a thick *biological shield*. The reactor system and steam generators are normally enclosed in a *containment vessel*. In case of a potentially dangerous malfunction in the reactor the containment is sealed off, thus presenting a barrier against escape of radioactive material to the surroundings. Air and water effluents from the reactor station during normal operation are monitored and (when prescribed activity levels are exceeded) purified from radioactive contaminants.

The fuel elements can only be used to much less than 100% consumption of the ^{235}U originally present. Fission leads to the production of fission products, which accumulate in the fuel. Some of the fission products have very high neutron capture cross sections ($\gg 100$ b) and compete with the fission chain reaction for the neutrons. Before the reactor becomes "poisoned" by these fission products, part of the fuel elements has to be replaced. In case of LWR:s the reactor is shut down, the top of the pressure vessel removed and some of the older fuel elements replaced by new ones. The replacement is done by means of a fuel *charging* (discharging) *machine* (a few other reactor types permit fuel replacement during operation). Because of the large amounts of radioactive fission products, the used fuel elements are always allowed to "cool" (with respect to both radioactivity and heat) for months, often years, in water-filled *storage pools* located inside the containment (see Fig. 19.16).

The used fuel elements may later be *reprocessed* to recover the remaining amount of fissile material as well as any *fertile material* or regarded as waste; fertile atoms are those

which can be transformed into fissile ones, i.e. ^{232}Th and ^{238}U, which through neutron capture and β-decays form fissile ^{233}U and ^{239}Pu, respectively. The chemical reprocessing removes the fission products and actinides other than U and Pu. Some of the removed elements might be valuable enough to be isolated although this is seldom done. The mixed fission products and waste actinides are stored as *radioactive waste*. The recovered fissile materials may be re-fabricated (the U may require re-enrichment) into new elements for reuse. This "back-end" of the nuclear fuel cycle is discussed in Chapter 21.

19.2. ENERGY RELEASE IN FISSION

From Table 11.1 it is seen that in thermal neutron fission of ^{235}U the fission fragments are released with a kinetic energy of ∼165 MeV (on the average), the 2.5 *prompt neutrons* have an average kinetic energy of ∼5 MeV together, and the *prompt γ-rays* have an average of 7 MeV. This *prompt energy release* of ∼177 MeV is absorbed in the surrounding material.

The fission products are radioactive and decay through emission of β^-, γ, and X-rays; their total amount of decay energy is ∼23 MeV. About 10 MeV (the value is uncertain) escapes the reactor as radiation, and ∼1 MeV of the decay energy remains as undecayed fission products in the spent fuel when unloaded from the reactor; thus ∼12 MeV βγ decay energy (divided about equally between $\bar{E}_\beta$ and E_γ) is absorbed in the reactor. The neutrons not consumed by fission are captured in the reactor material with release of binding energy; it is estimated that this amounts to about 10 MeV. Thus the total amount of energy expended per fission in a shielded controlled reactor is about $177 + 12 + 10 \simeq 199$ MeV. The total energy release varies only a few MeV between the different fissile nuclei. As a practical average value 200 MeV per fission can be used regardless of the fissioning nucleus:

$$E_f \approx 200 \text{ MeV per fission} = 3.20 \times 10^{-11} \text{ J per fission} \tag{19.2}$$

Thus $(3.20 \times 10^{-11})^{-1} \approx 3.1 \times 10^{10}$ fissions s^{-1} correspond to the production of 1 W of reactor heat. The heat power of a reactor can be written

$$P = E_f \mathrm{d}N_f/\mathrm{d}t \tag{19.3}$$

where $\mathrm{d}N_f/\mathrm{d}t$ is the number of fissions per second. A nuclear power station producing 3 GW heat (GW_{th}; th for thermal) has an electric output of 1 GW_e at a 33% *efficiency* in converting the thermal energy into electric; this corresponds to 8.1×10^{24} fissions d^{-1}. Since the weight of a ^{235}U atom is $M/\mathbf{N}_A = 3.90 \times 10^{-25}$ kg, this would correspond to the fission of $8.1 \times 10^{24} \times 3.9 \times 10^{-25} = 3.16$ kg ^{235}U d^{-1}; the real consumption of ^{235}U is slightly different and varies with time as discussed in §19.9.